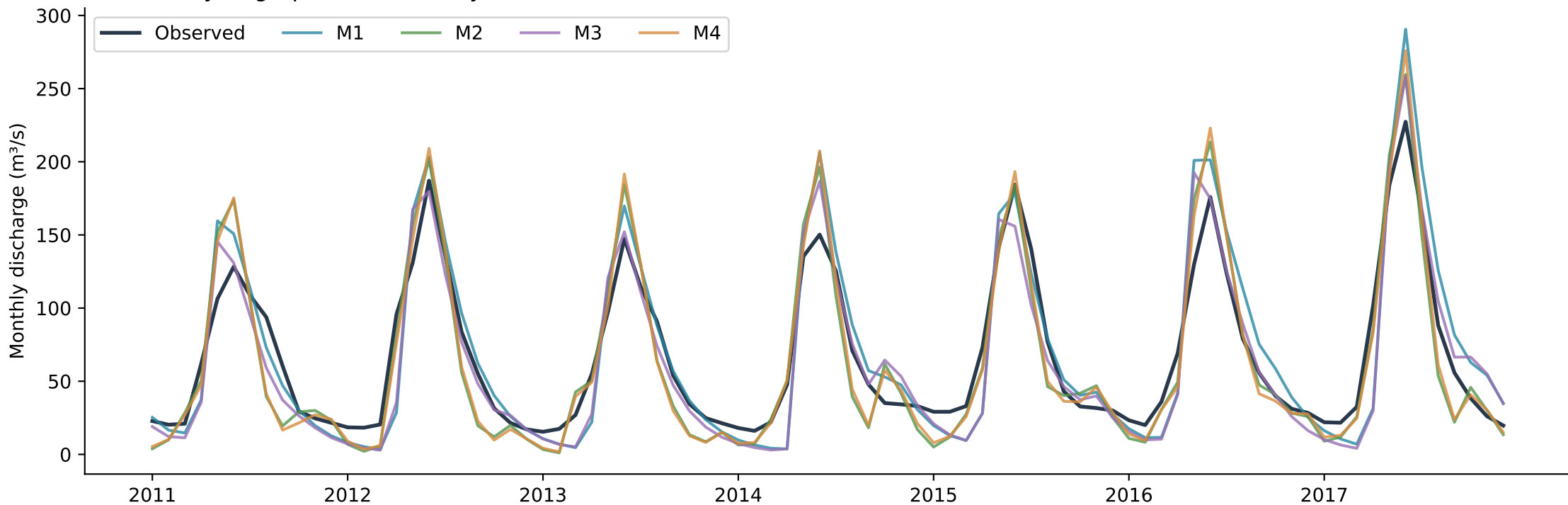
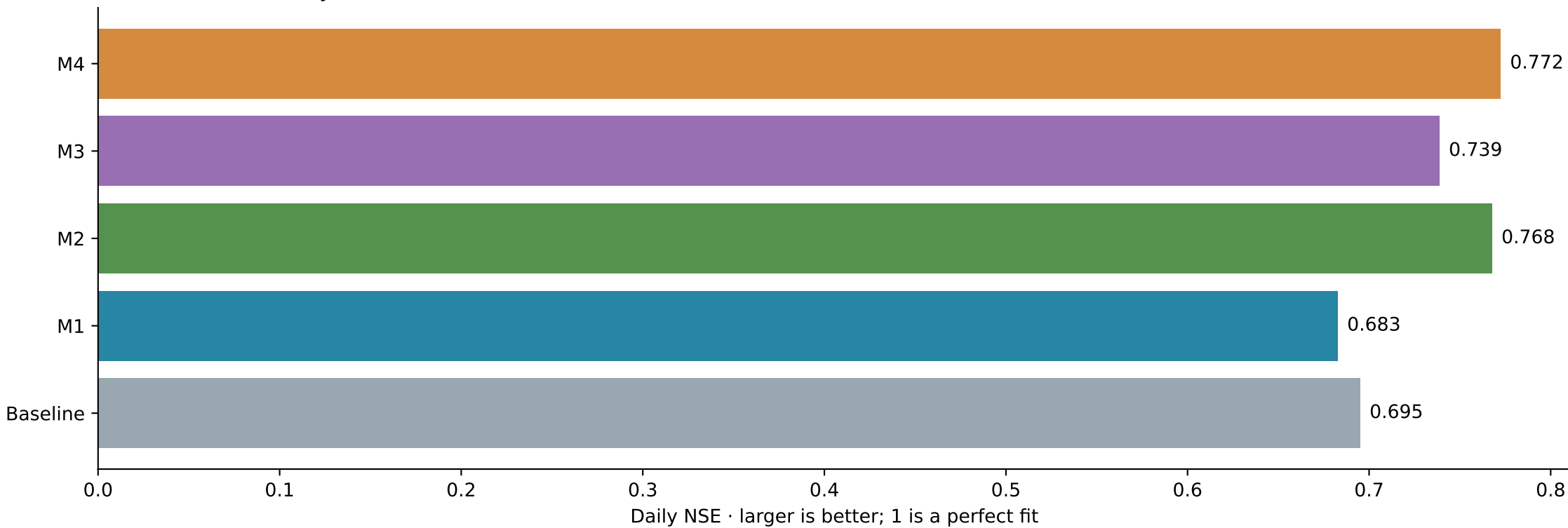


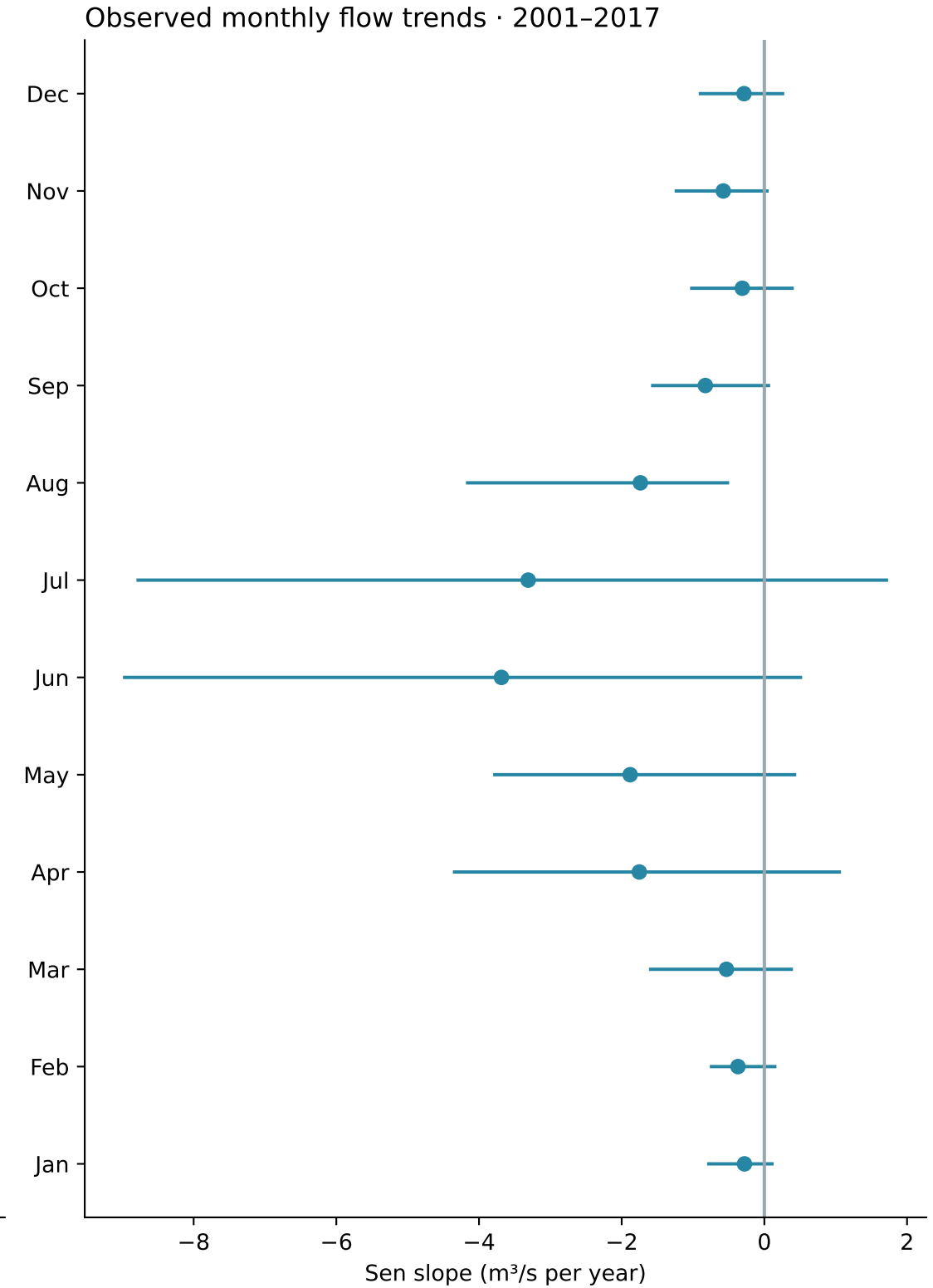
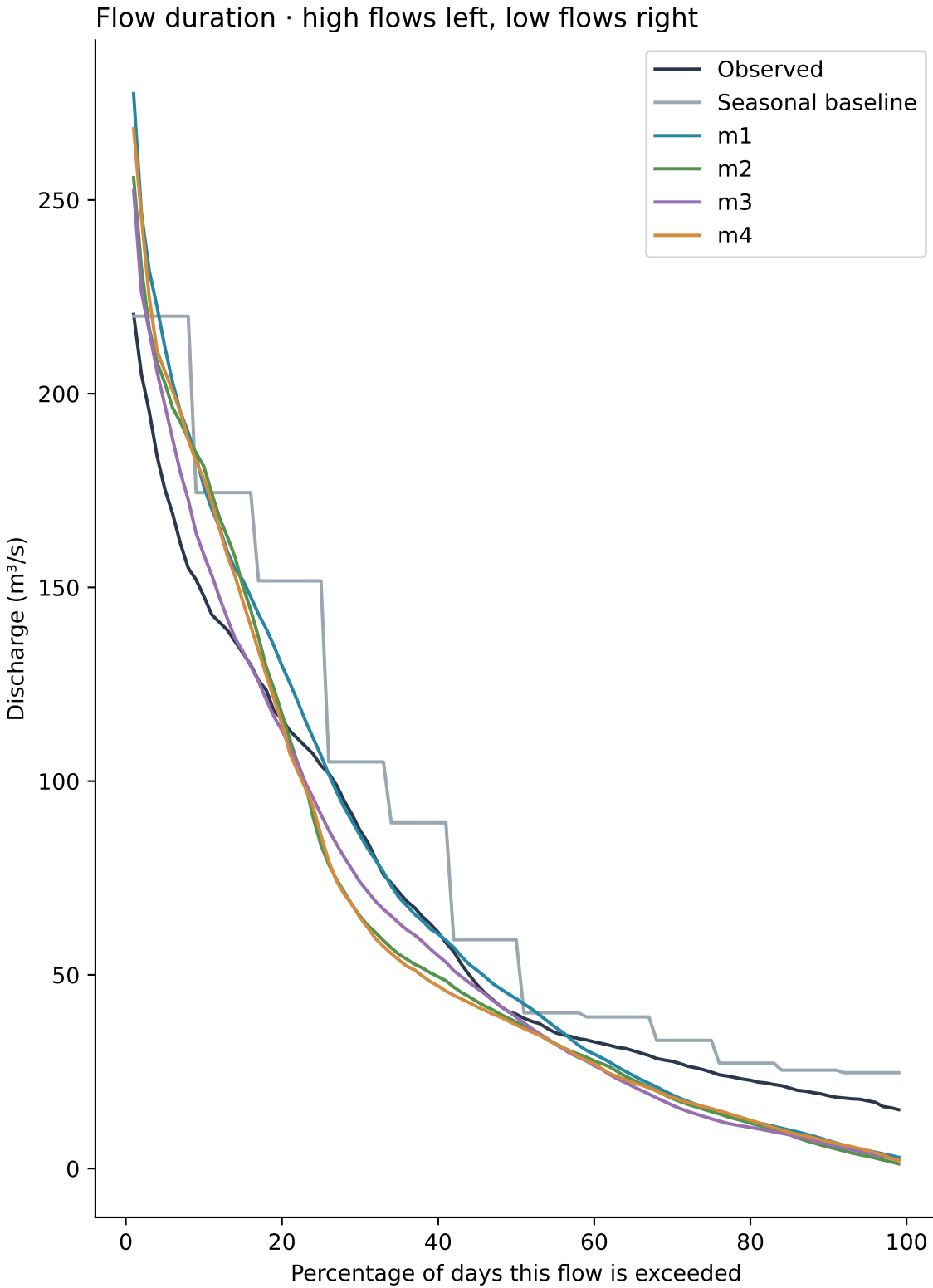
Held-out hydrograph · all four daily model structures



Same screened daily observations · 2011-2017

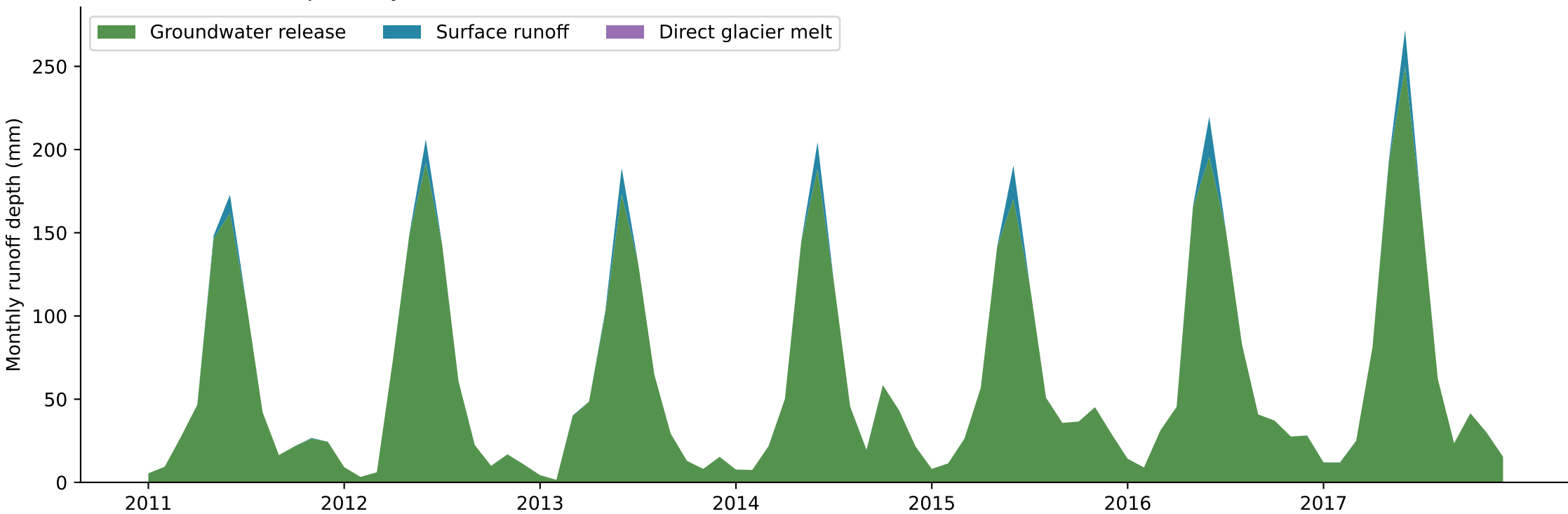


Calibration: 2001-2010; warm-up: 2000. Adaptations with ERA5 weather and a provisional drainage boundary; not thesis score replication.

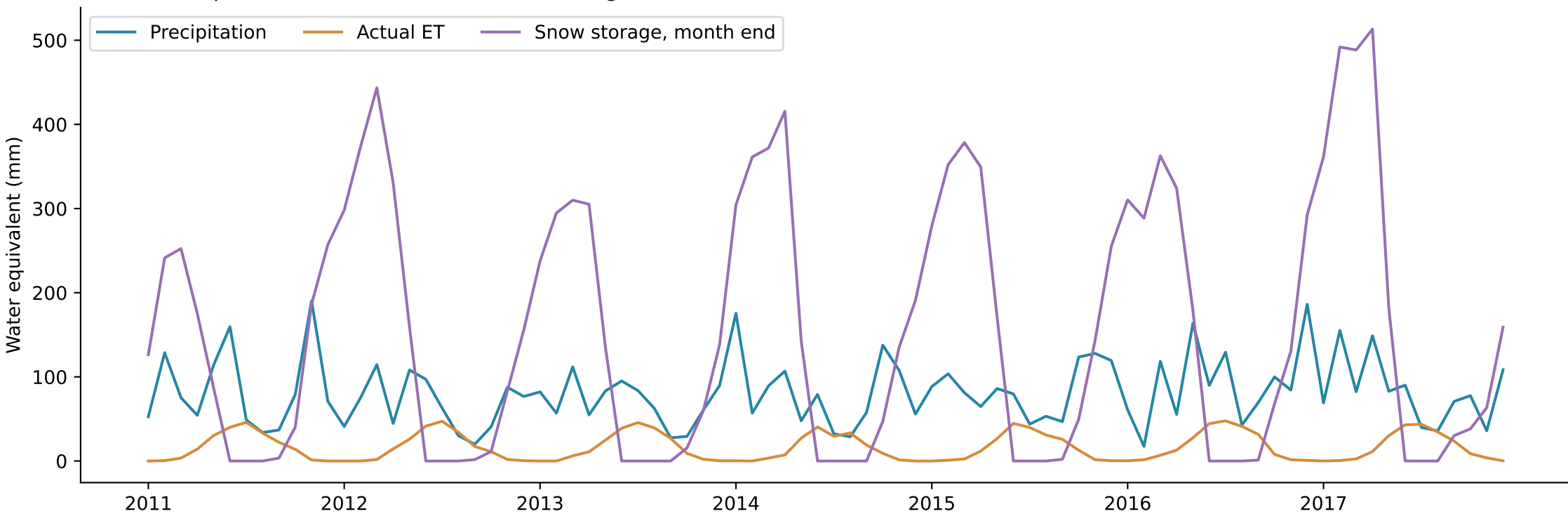


Flow duration uses screened held-out days. Trend intervals are nominal; serial dependence remains, with twelve-test Holm p-values in the workbook.

M4 modelled runoff pathways

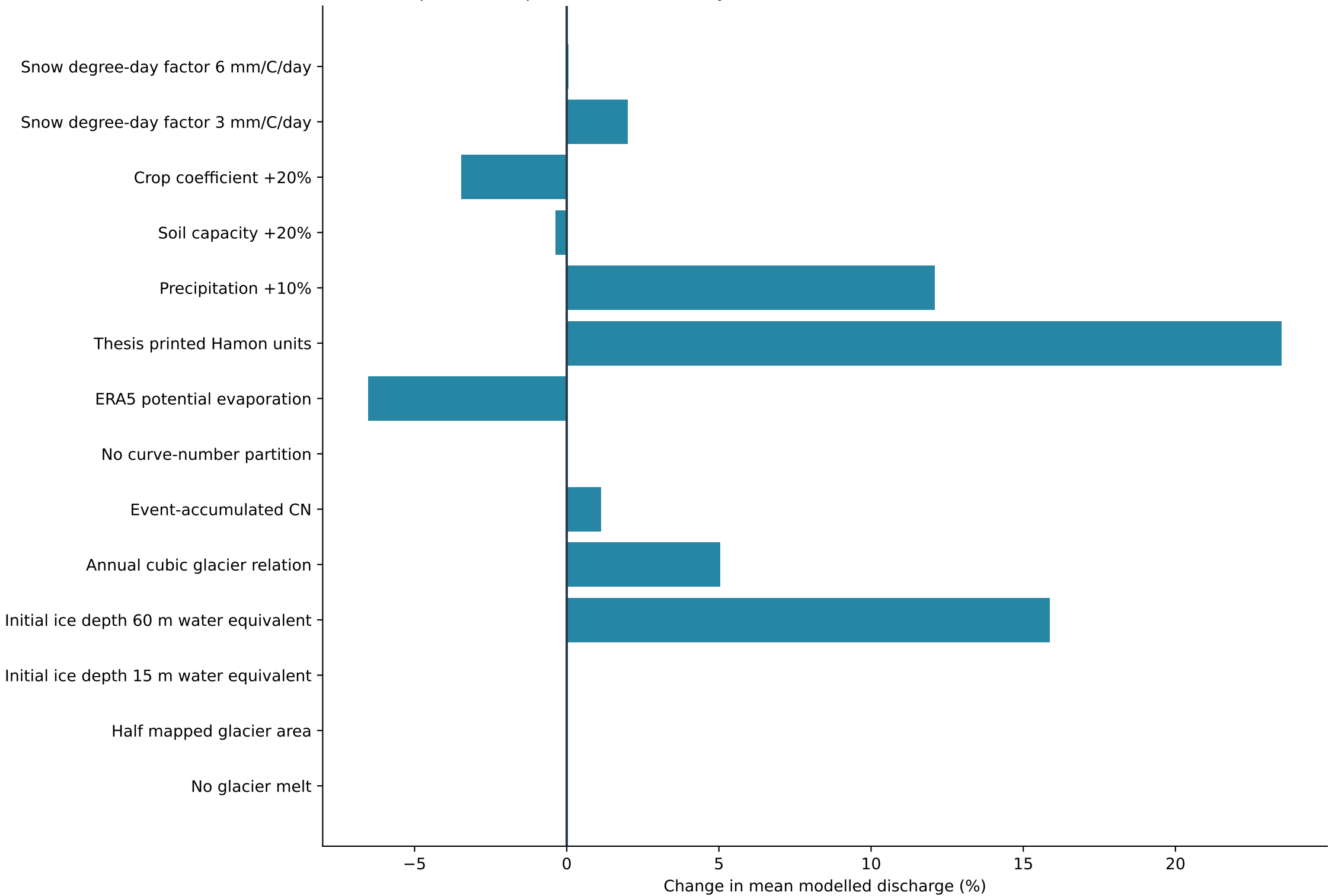


M4 water inputs, loss and seasonal snow storage

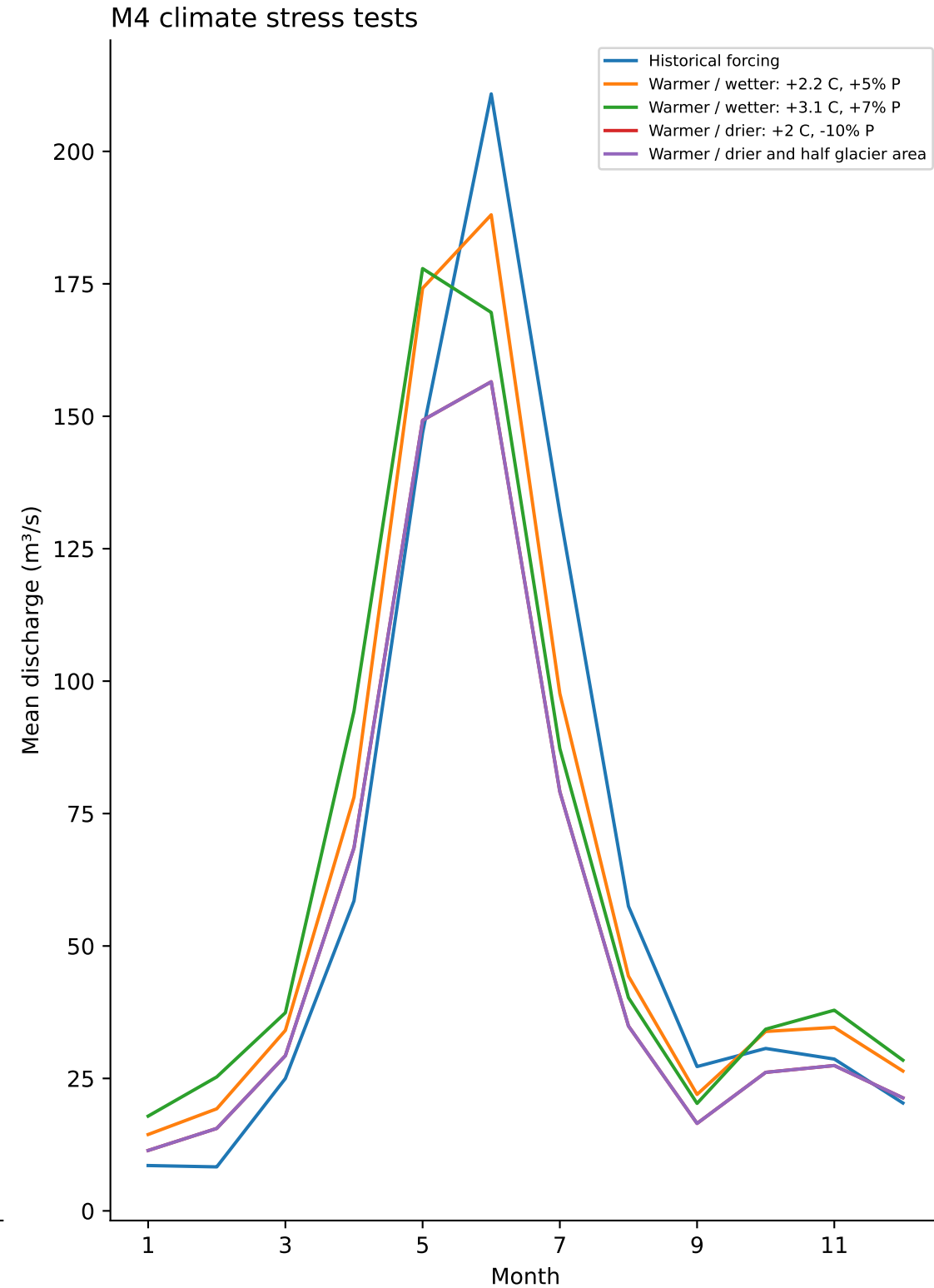
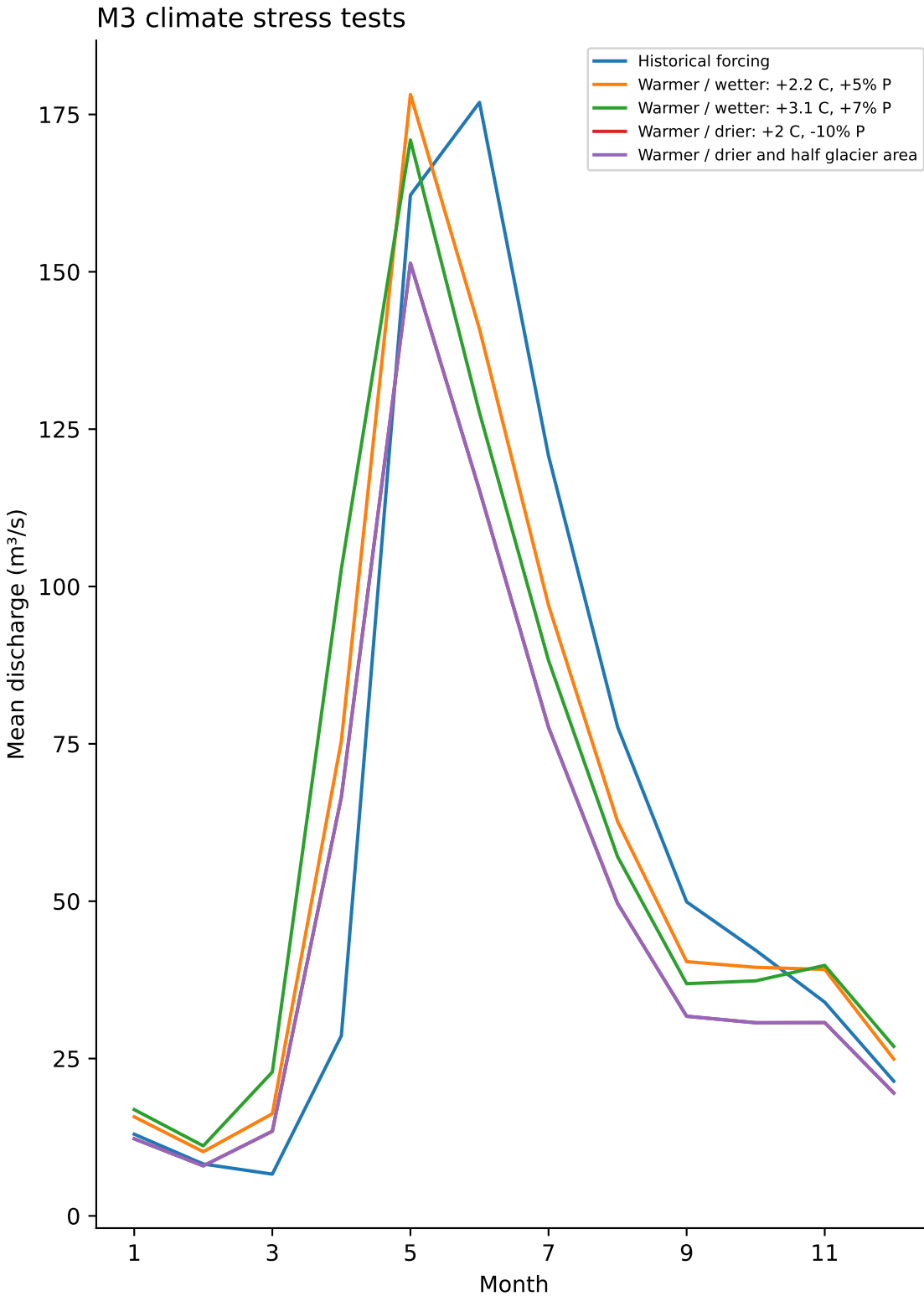


Snowmelt feeds surface/groundwater paths and is not added twice. Glacier contribution is a fixed-inventory model scenario, not observed attribution.

M4 assumptions and parameter sensitivity · frozen calibration

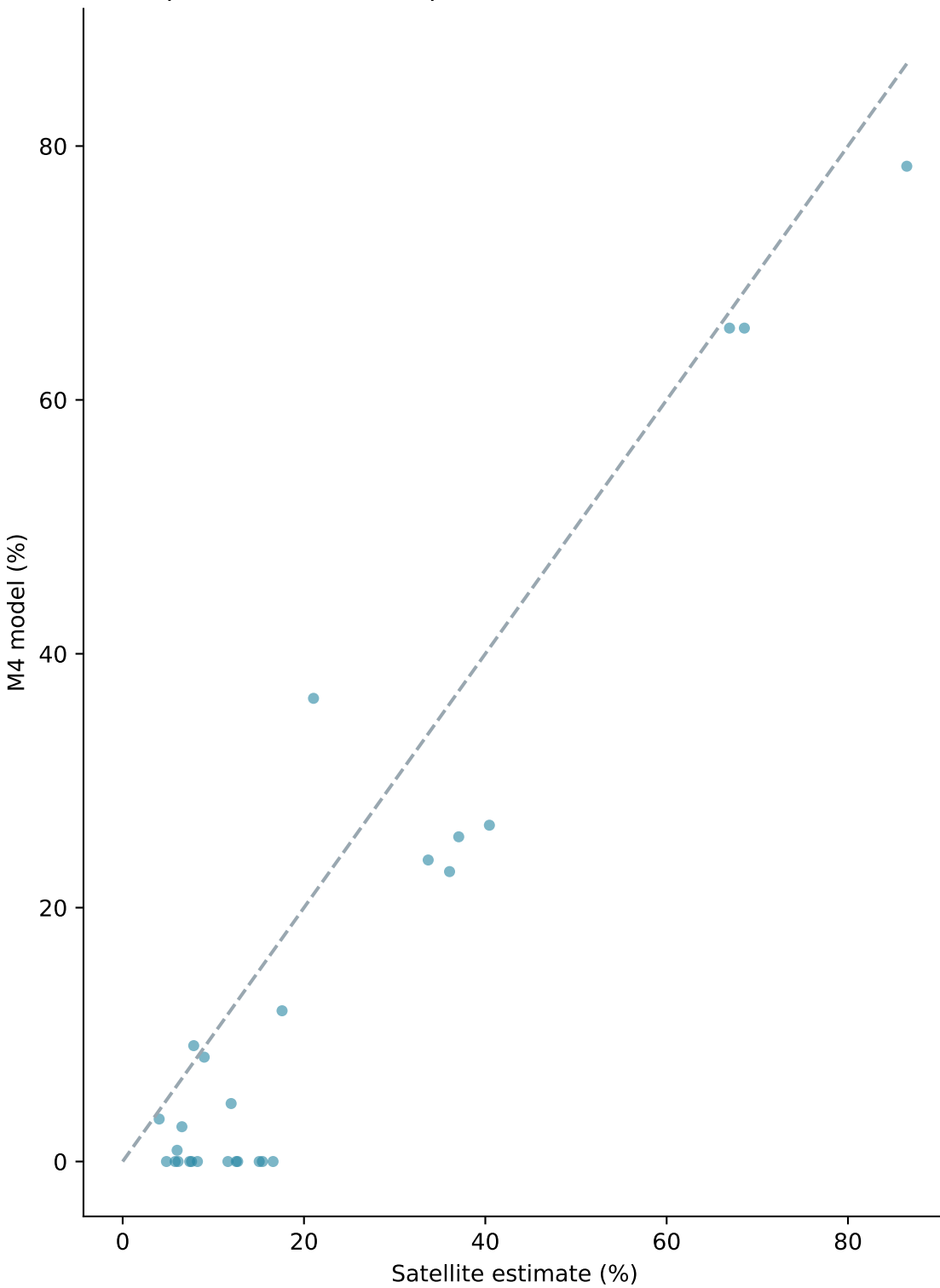


One-at-a-time perturbations; not causal variable importance or confidence intervals. RMSE changes and conservation checks accompany each scenario.

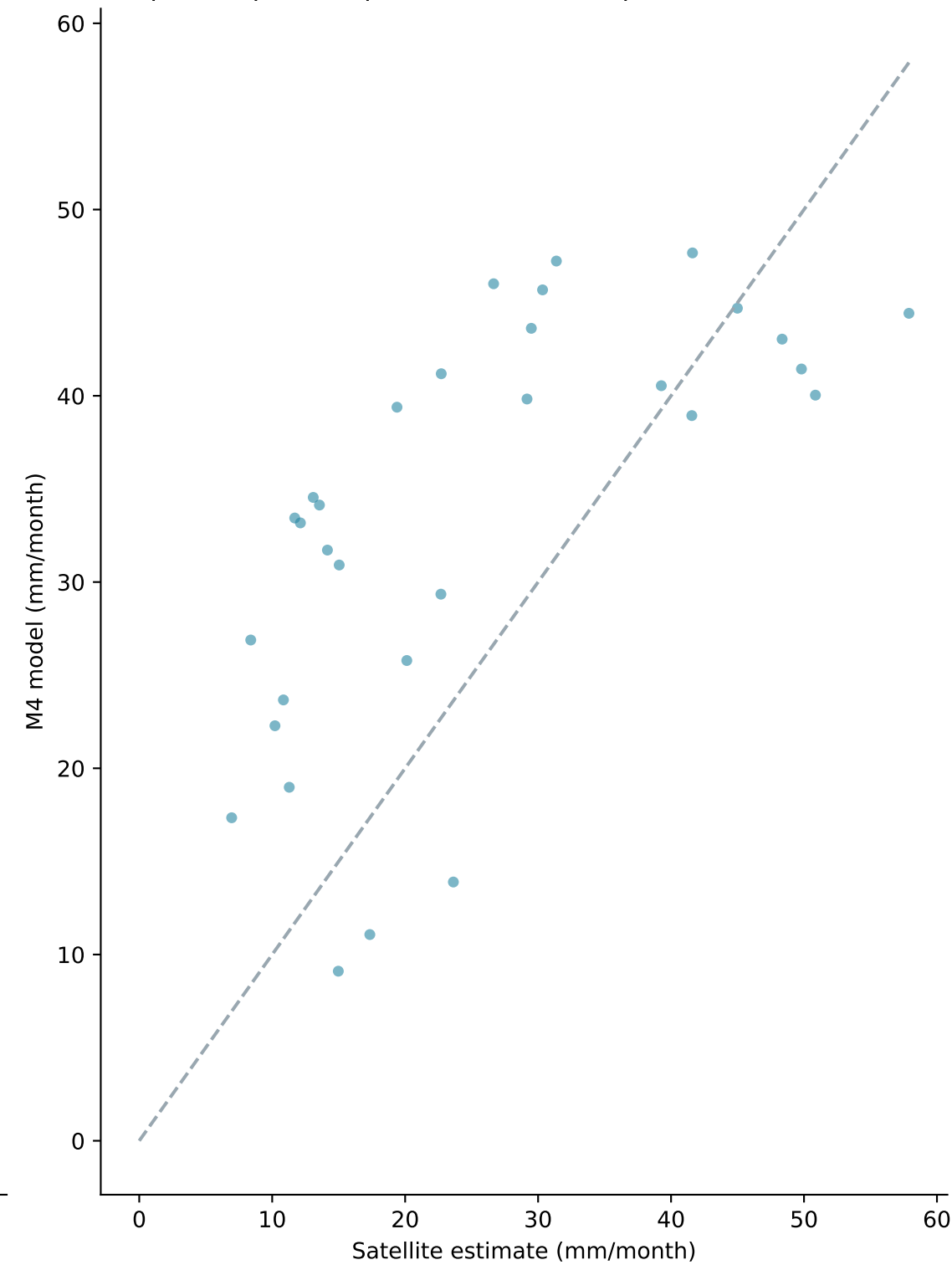


Historical daily sequence perturbed with frozen parameters. These are stress tests, not calibrated 2050 projections or scenario probabilities.

Snow process check · 27 pairs



Evapotranspiration process check · 31 pairs



2011-2017; coverage thresholds retained. Three-zone snow proxy and whole-basin AET have different spatial support from satellite valid pixels.